

A two-dimensional nanochannel facilitates ionic conductivity of a deep eutectic solvent for an efficient supercapacitor

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ABSTRACT

As an environmentally benign promising emerging class of versatile solvents, deep eutectic solvents (DESs) are inexpensive, non-corrosive, compatible with electrodes and eco-friendly, and attractive for cost-efficient energy storage systems. However, the limited ionic conductivity and liquid state constrict the wide application of bulk DESs in energy storage. Herein, we present a nanoconfined system by confining a common DES, choline chloride-ethylene glycol, into the two-dimensional nanochannel of graphene oxide (GO) membranes. Under infusing DES into GO nanochannels, the ionic conductivity of DESs has increased 64 times as that of the bulk one. The molar ratio and temperature influence of DES components on the ionic conductivity of nanoconfined GO-DES membranes have been investigated. As a proof of conception, the resulting GO-DES membranes were utilized as quasi-solid liquid electrolytes for supercapacitors. The GO-DES-based supercapacitor demonstrates a 50.88% energy density improvement than bulk one. The nanoconfined system of GO-DES10 exhibits high specific capacitance of 120 F g⁻¹ at 1 A g⁻¹ and an energy density of 10.23 Wh kg⁻¹ at power density of 1 kW kg⁻¹. These are the best performance among similar ionic liquid supercapacitors. This enhanced ionic conductivity in nanoconfined DESs will offer a new way for DESs in the electrochemical area.

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1. Introduction

The recent increase in electricity and fuel prices caused by the tension in the world's energy supply has bolstered the urgent need to develop new eco-friendly, pollution-free, and dependable energy storage systems. Growing attention has been paid to the development of energy storage technology, particularly supercapacitors [1]. It is crucial to develop green and reliable electrolytes with appropriate conductivity in energy storage equipment [2–4]. Common electrolytes for supercapacitors are usually classified into three species: aqueous electrolyte (KCl [5], H₂SO₄ [6,7], KOH [6,7], and

Na₂SO₄ [8]), organic electrolyte (ethylene carbonate [9], dimethyl carbonate [10], and acetonitrile [11]), and ionic liquid (EMIMBF₄ [12] and EMI-TFSI [13]). Aqueous electrolytes are limited due to the relatively low decomposition potential of water [14]. Meanwhile, organic electrolytes are mostly toxic and may pose many safety risks to the environment. Compared with aqueous and organic electrolytes, ionic liquids show unique advantages of much higher conductivity and a high voltage range [15]. However, they are generally so expensive and sometimes need to be mixed with organic solutions to achieve maximum effectiveness.

Deep eutectic solvents (DESs), as room-temperature ionic liquid analogs, are comprised of a variety of hydrogen bond acceptors and hydrogen bond donors [16,17]. One of the most attractive advantages of DESs is that their properties could be fine-tuned by changing the components and the molar ratio of hydrogen bond acceptors to hydrogen bond donors. Compared with ionic liquids, a DES is more suitable for electrolytes due to its lower price, easier preparation, and biodegradability. Attributing to the above advantages, many reports have utilized DESs in electrochemical storage applications in recent years. Zhang's group reported a novel DES

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electrolyte for supercapacitors simply by mixing choline chloride (ChCl) and ethylene glycol (EG). This DES-based electrolyte can operate normally over a wide temperature range from 40 to 115 °C [18]. A solid polymer electrolyte based on a ChCl/lactic acid DES entrapped in a chitosan matrix as a potential separator for electrochemical energy storage devices had been developed [19]. Sara Azmi found a new approach to the DES application of ChCl–urea, commonly called reline, and its aqueous mixtures as eco-friendly, affordable, and anticorrosive electrolytes for carbon-based electrochemical capacitors [20]. Cui et al. constructed a solid Zn^{2+} -ion conductor by eutectic liquids. The high-entropy eutectic-networks weaken the ion-association and form interfacial Zn^{2+} -percolated channels on the nucleator surfaces, resulting in a solid crystal with exceptional selectivity for Zn^{2+} transport [21]. All these reports confirm the wide application prospect of DESs in electrochemical storage systems.

However, the limited ionic conductivity restricts the further development of DESs in the field of the electrochemical energy storage. Meanwhile, a liquid electrolyte has the risk of thermal runaway and low volume energy density, which limits their application in the electrochemical energy storage. Layered two-dimensional (2D) materials have attracted extensive attention due to their facile preparation methods and excellent performance of 2D nanochannels [22–24]. In the quest to develop energy storage with both high power and high energy densities, while maintaining a high volumetric capacity, recent results show that a variety of laminated 2D materials exhibit rapid kinetics of ion transport by acting as nanofluidic platforms [25–27]. 2D materials provide an extended interlayer space that can hold more ions and thus improve the flux. The high ionic conductivity is attributed to the confining effect of 2D materials to form the ordered ion transport channels in an interlayer [23,24].

Here, this work focuses on exploring the mechanism of greatly enhanced ionic conductivity of a common nanoconfined DES with some experimental facts. GO nanosheets and ChCl-EG DES are selected here to investigate the nanoconfinement effect on ionic conductivity. The transport properties of different molar ratios of ChCl-EG at different temperatures have been investigated, accordingly. Moreover, characterizations were applied to confirm the existence of a hydrogen bond and the effect of the nanoconfinement of DESs. The ionic conductivity increments of the DES involved in this work are more than a factor of 64, significantly improving the performance of DESs as electrolytes. Meanwhile, the GO-DES-based supercapacitor exhibits a 30.86% increase compared with the bulk one in specific capacitance. Furthermore, the GO-DES-based supercapacitor demonstrates a 50.88% improvement than the bulk one in energy density, which offers a more profound application of the conductive properties of the DES in 2D nanoconfinement. The excellent electrochemical performance of the nanoconfined system makes the GO-confined DESs have broad application prospects in electrochemical energy storage.

2. Experimental section

2.1. Chemicals and materials

ChCl (AR, purchased from Aladdin), EG (99.8%, purchased from Aladdin). Anodic aluminum oxide (AAO) membranes were used as substrates for the filtration. Graphite powder was purchased from Alfa Aesar. Phosphorus pentoxide (P_2O_5 , $\geq 98.5\%$) and potassium persulfate ($\text{K}_2\text{S}_2\text{O}_8$, $\geq 99.5\%$) were purchased from Sigma-Aldrich. Concentrated sulfuric acid (H_2SO_4), potassium permanganate (K_2MnO_4), hydrogen peroxide (H_2O_2), and hydrochloric acid were purchased from Sinopharm Chemical Reagent Co., Ltd. Ultrapure

water of $18.2 \text{ M}\Omega \text{ cm}^{-1}$ was produced by a Millipore direct-Q system (Millipore, U.S.A). Activated carbon (AC), carbon sheet (purchased from Nanjing Xianfeng Nano Material Technology Co., Ltd.), carbon black, and N-Methylpyrrolidone.

2.2. Synthesis of ChCl-EG DES

ChCl and EG with a specific molar ratio were poured into a well-sealed vial. Then the mixture was agitated with a magnetic stirrer at 70 °C for approximately half an hour at 5000 rpm until a transparent homogenous liquid was obtained.

2.3. Synthesis of GO-DES membrane and nanofluidic devices

The GO nanosheets were prepared by the modified Hummers' method [28]. An amount of 5 mL of the GO dispersion was used to fabricate GO on AAO substrates by vacuum filtration. GO membranes can be separated from AAO easily to get self-standing membranes. And GO was dried overnight under a vacuum to remove water. The GO-DES membranes were fabricated by dropping 100 μL of the prepared DES solution on the GO membrane for 30 min and then spinning-coated at 5000 rpm for 60 s. The GO membrane was cut into rectangular pieces and immersed in a blend of a polydimethylsiloxane (PDMS) precursor and curing agent (20:1). After curing, two reservoirs were carved out in the PDMS device to expose the two ends of the membrane to store the ChCl-EG electrolyte.

2.4. Assembly of supercapacitors of DES and GO-DES

The electrochemical experiments in a two-electrode system were performed in a CR2032 coin-type cell. The AC, conductive carbon black, and polyvinylidene fluoride of 8:1:1 were added to N-Methylpyrrolidone and mixed in a sealed beaker and magnetic stirred for 24 h. Then, the obtained slurry was uniformly coated onto a carbon sheet and dried at 120 °C for 12 h to prepare the working/counter electrode. A glass fiber membrane was used as the separator for the symmetric bulk DES supercapacitor. GO dispersion was vacuum filtrated on a glass fiber membrane and then dried in an oven at 120 °C for 4 h to prepare the separator of the GO-DES supercapacitor.

2.5. Characterization

A sigma 300 field-emission scanning electron microscope equipped with an energy dispersive X-ray spectrometer was employed to obtain scanning electron microscope images and energy dispersive X-ray mapping images of GO and GO-DES. Shimadzu XRD-6000 with a $\text{Cu K}\alpha$ radiation source was used to get X-ray diffraction patterns of GO and GO-DES. A contact angle was performed by a Data-Physics OCA-20 video-based contact angle measuring device. Fourier transform infrared reflection spectra of the liquid and film samples were measured by a Nicolet 5700 infrared spectrometer of Thermoelectric Company of America, which is used to analyze the surface functional groups of liquid solvents and film samples. The differential scanning calorimetry (DSC) instrument used in this project is a reactive differential scanning calorimeter from NETZSCH, Germany. The model is DSC 204HP, the test temperature range is -150 – 80 °C, and the temperature changing rate is 10 °C/min. The viscosity of the DES was measured with a capillary viscometer at room temperature. Three capillaries of an internal diameter of 1.0, 1.5, and 2.5 mm were employed to perform the measurements for the range of viscosity of interest. The viscometer was calibrated with the standards provided by the manufacturer. The pulsed-field gradient nuclear

magnetic resonance (PFG NMR) measurements were obtained on a 700 MHz nuclear magnetic resonance spectrometer (AVANCE NEO, Bruker company). The sweeping range of ^1H is 0–7 ppm.

2.6. Electrochemical characterization

The property of the ionic transport was obtained by a two-electrode system on an electrochemical workstation (CHI-660D, Shanghai ChenHua instruments Co., Ltd, China). The ionic conductivity of the bulk solution was tested by a conductivity meter (DDS-307A). The cyclic voltammetry (CV) and galvanostatic charge and discharge (GCD) measurements were tested in the potential range of 0–2 V at different scanning rates and current densities. Electrochemical impedance spectroscopy (EIS) plots were obtained from 0.01 Hz to 100 kHz. The Nyquist plots reveal the transport of electrolyte ions between the electrodes, where the charge transfer resistance can be determined by a semicircular diameter (R_{ct}). The intersection point is referred to equivalent series resistance (R_s), including the resistance of the electrode, connection resistance, and electrolyte of the device. The diameter of the semicircle is related to the charge transfer resistance, which includes the electrode–electrolyte interface and the pore structure of the electrodes. Z-view software was used to obtain the resistance values of the measured electrochemical systems. The voltage window of supercapacitors was determined by the previous study of 0–2 V [18]. The cycling performance and charge-discharge curves were measured by Land CT2001 A.

The ionic conductivity of the nanoconfined DES can be calculated by equation (1). The strips of GO membranes were embedded in a PDMS elastomer with two reservoirs containing a volume of about 0.1 mL DES. Two Pt wires were inserted into the reservoirs and served as electrodes to explore the electrochemical performance.

$$\sigma = \frac{l}{R \times S} \quad (1)$$

Where, σ is the ionic conductivity, l is the distance between the two Pt wires, R is the resistance which can be obtained from the Nyquist plots, and S is the contact area.

The ionic conductivity against different temperatures exhibits well-fitting Arrhenius-type dependence (equation (2)), from which the activation energy can be calculated

$$\ln \sigma = \ln \sigma_0 - \frac{E_a}{RT} \quad (2)$$

Where, σ is the conductivity, σ_0 is a constant, and E_a is the activation energy.

According to CV and GCD results, the mass specific capacitance (C), energy density (E), and power density (P) of supercapacitors are calculated by the following equations (3)–(5):

$$C = \frac{1}{U \cdot v} \int i dV \quad (3)$$

$$E = \frac{0.5CV^2}{3.6} \quad (4)$$

$$P = \frac{3600 \times E}{t} \quad (5)$$

Where U is the potential window, v is the scan rate, i is the discharge current, and dV is the infinitesimal changes in potential.

3. Results and discussion

3.1. Membrane preparation and characterization

Fig. 1 illustrates the synthesis procedure of the GO-DES membrane as described in the experimental section. To investigate the ionic transport behaviors of DESs in a 2D nanoconfined channel, GO was selected to construct the nanometric channel. Different molar ratios of ChCl:EG (1:2, 1:4, 1:6, 1:8, and 1:10) were tested and they were named DES2, DES4, DES6, DES8, and DES10. The corresponding digital photos of these DES solutions are shown in Fig. 2a. Generally, ChCl-EG DES is formed by the combination of two constituents generating a new liquid phase containing the Cl^- -EG hydrogen bond network. GO nanosheets were fabricated by the common modified Hummers' method [28]. There are obvious wrinkles on the GO surface (Fig. S1). But after being filled by a DES, the wrinkles are less (Fig. 2b). The GO-DES was prepared with a 5 mg GO solution, and the film thickness is 4.8 μm (Fig. 2c). The thickness of GO increased about twice as much as before (Fig. S1) DESs penetrated into it. The contact angle of the DES on the GO surface is about 35° (the inset image in Fig. 2b), showing good wettability of the DES on the GO membrane. The energy dispersive X-ray spectroscopy of Cl and N of ChCl-EG in Fig. 2d prove that ChCl and EG have been filled into the interlayered nanochannels of GO. The intrinsic peak position of GO (1 mg/mL, 5 mL) is about 10.68° , and the resulting layer spacing is 0.82 nm (Fig. 2e). The X-ray diffraction peak position of the GO-DES is about 4.38° , and the layer spacing is 2.01 nm. The change of the layer spacing before and after the introduction of DESs is consistent with the change of the film thickness, which also proves that the DES fully and uniformly fill the nanochannel of GO [29]. The infrared spectra of the GO, DES, and GO-DES and the corresponding characteristic vibration peak positions (Fig. 2f) are the same as reported in the literature [30,31]. For the DES, the broad band between 3200 and 3600 cm^{-1} is the stretching vibration peak of an O–H bond, which can prove the existence of hydrogen bonds in DESs [30,31]. The 3031 cm^{-1} peak is the N–H stretching vibration mode in ChCl. The peaks at 2940 and 2875 cm^{-1} correspond to the antisymmetric and symmetric stretching vibrations of sp^3 hybrid C–H bonds in EG, respectively. The vibration band at 1481 cm^{-1} is the bending vibration of CH_2 of alkyl in ChCl, and 863 cm^{-1} is the symmetric expansion of C–N^+ . The peaks at 1087, 1041, and 883 cm^{-1} correspond to C–O stretching, C–C–O asymmetric stretching, and C–C–O symmetric stretching vibrations in EG, respectively. The vibration band of the DES in the GO-DES is generally consistent with that of a pure DES, but some peak positions have some deviations. In the GO-DES, the O–H stretching vibration and sp^3 hybrid C–H bond stretching vibration have blue shifts compared with a pure DES, which indicates that the hydrogen bond state in the nanoconfined DES has changed. Since the stronger the hydrogen bond, the smaller the corresponding vibration peak [32–34], it can infer that oxygen-containing functional groups on the surface of GO nanosheets can easily form hydrogen bonds with Cl^- of a DES and active sites in EG. The interaction between the GO layer and DES causes partial destruction and reconstruction of the local hydrogen bond network near the GO layer in the DES. The final result is to weaken the hydrogen bond strength in the DES to a certain extent. Thermodynamic properties of nanoconfined DESs were investigated by DSC (Fig. 2g). The hydrogen bond between the GO layer and DES also affects the thermodynamic properties of the nanoconfined DES. The melting point of the bulk DES is -33.2°C . The hydrogen bond attraction of GO nanosheets to Cl^- and EG induces Cl^- rich and EG rich layers near the GO layer, while the CH^+ ions with larger molecular size are relatively more stable. This compositional redistribution in the nanospace results in Cl^- and EG delamination.

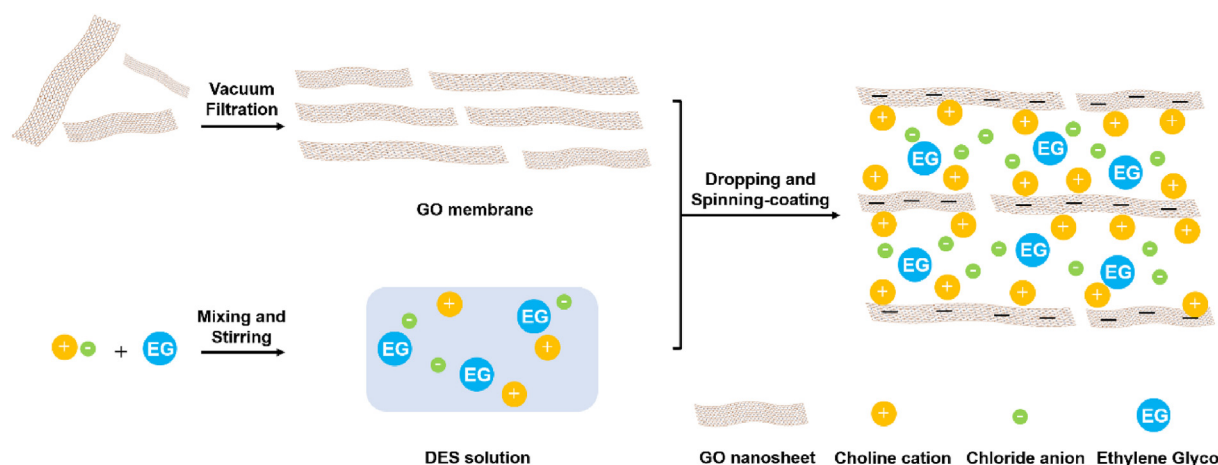


Fig. 1. scheme of the synthesis process of the GO-DES. Choline chloride and ethylene glycol were mixed by stirring at 70 °C to prepare the DES solution. GOs were fabricated by vacuum-filtrating GO dispersion. And the DES was immobilized into GO by simple dropping and spinning-coating methods. DES, deep eutectic solvents; GO, graphene oxide.

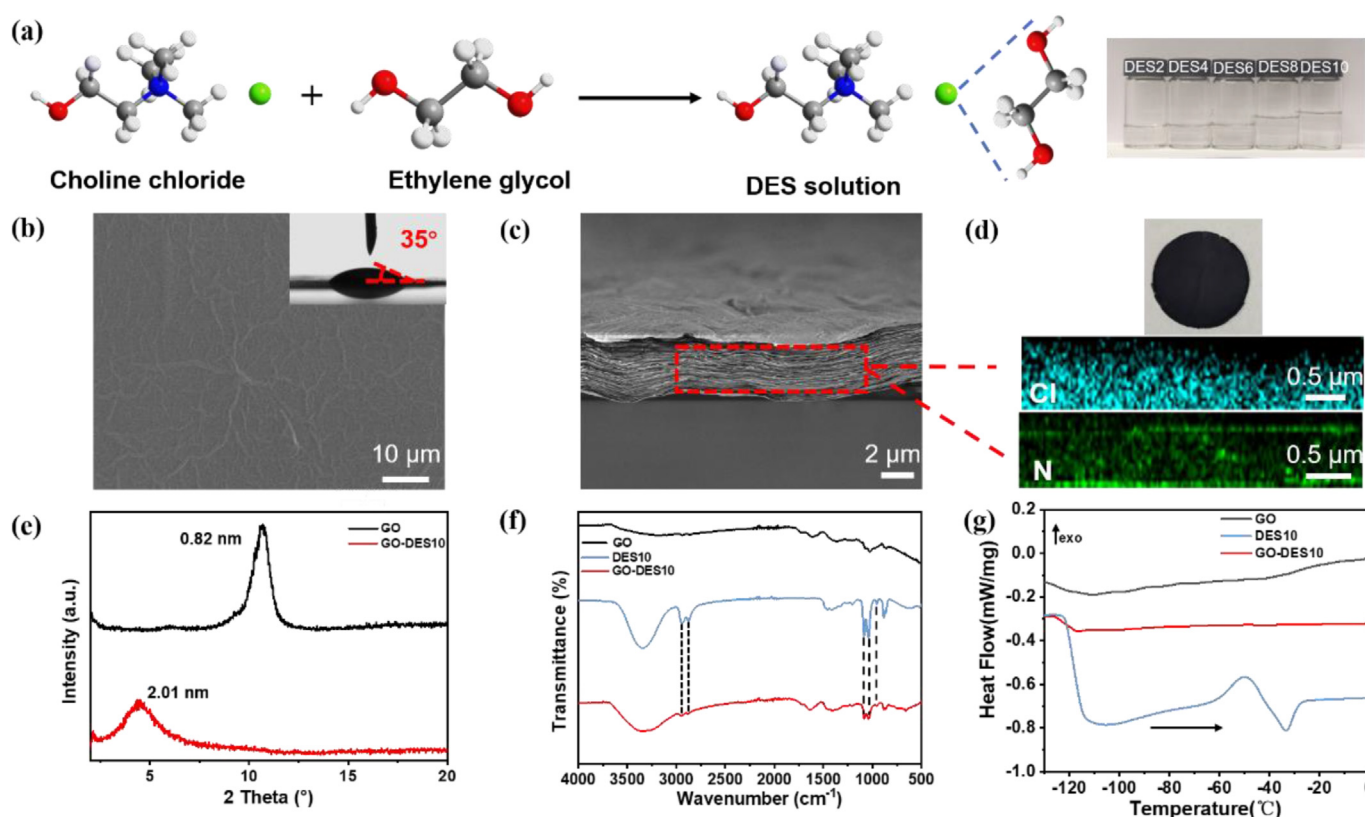


Fig. 2. (a) The molecular representation and synthesis of the ChCl-EG DES, the digital photo of the DES solutions. (b) The surfaces SEM image of GO-DES membranes. The inserted image is the contact angle of the DES on the GO membrane. (c) The cross-section SEM image of the GO-DES. (d) The digital photo of the GO-DES membrane, EDS mapping images of Cl and N elements recorded from the marked zone of Fig. c, respectively. (e) XRD patterns of GO, GO-DES membranes. (f) FTIR spectra of GO, DES, GO-DES membranes. (g) DSC patterns of the DES and GO-DES membrane. DES, deep eutectic solvents; GO, graphene oxide; SEM, scanning electron microscope; XRD, X-ray diffraction; FTIR, Fourier transform infrared; EDS, energy dispersive X-ray spectroscopy; DSC, differential scanning calorimetry; EG, ethylene glycol; ChCl, choline chloride.

Due to the redistribution of anions, the electrostatic force between Ch^+ and Cl^- is further pulled and weakened, resulting in a greater degree of charge delocalization and a lower melting point [29]. To identify the nature of surface chemistry in GO and the GO-DES, an XPS measurement was further employed. C 1s and O 1s XPS display the major species of GO membranes. Wide scan spectra of GO-DES are shown in Fig. S1. Compared with GO (Fig. S1), the contents of N

1s and Cl 2p XPS prove that the DES is infused into the GO membrane (Table S1 and Table S2). As for the viscosity property, the viscosities of DES2 to DES10 are 42.60, 29.67, 23.10, 20.82, and 17.3 mPa s, respectively (Fig. S2). The increment of viscosity of the higher ratio of ChCl in the DES mostly attributes to the formation of hydrogen bonds, which results in lower molecular mobility [35].

3.2. Ionic transport of nanoconfined deep eutectic solvents

The ionic conductivity was measured by a nanofluidic device (Fig. 3a), which supports ions transport through 2D membranes. EIS tests were carried out to explore the conductivity of GO-DES systems (Fig. 3b). The bulk conductivity was measured by both the conductivity meter and EIS. The ionic conductivity of bulk DESs and their nanoconfined counterparts can be seen in Fig. 3c. There is no significant difference in the conductivity of different molar ratios of the bulk DES. On the contrary, with the decrease of the proportion of ChCl from 1:2 to 1:10 in the DES, the ionic conductivity of the nanoconfined system is constantly and significantly increasing. The conductivity of the confined system is higher than that of the bulk at all molar ratios (Fig. 3c). This is because under nanoconfinement, the anion and cation will be further dissociated as an ionic liquid-like electrolyte [23]. The ions will be orderly arranged to form faster ion channels [36], which leads to the increase of conductivity under a confined condition. Moreover, the viscosity of the DES was found to have an important influence on ion transport. Lower viscosity contributes to relatively higher conductivity due to high ion mobility [35]. The viscosity of the DES is mainly determined by the interaction between ChCl and EG.

The previous simulation was found when ChCl and EG were confined in GO nanosheets, with the decreasing of the ChCl molar ratio from 1:2 to 1:10, the interaction between ChCl and EG decreased with the increasing conductivity [37]. Therefore, the DES10 confined in the GO nanochannel with lowest viscosity exhibited the highest conductivity. Interestingly, when the molar ratio of ChCl:EG increased above 1:10, it can be seen that the conductivity of the nanoconfined system shows a downward trend. But it is still larger than the bulk one. This may be due to after the

hydrogen bond donor of the DES reaches a certain concentration, the excessive EG does not have enough ChCl as the receptor to interact with it. It destroys the balance between the hydrogen bond donor and the acceptor of the DES system, which is unfavorable for its ion transport. At the same time, the molecular volume of EG is relatively large, so the resistance is not conducive to its transport. It is interesting that all the corresponding conductivities of GO-DES membranes are higher than that of the bulk ones. The highest improvement of the GO-DES membrane with the ChCl:EG ratio of 10 is about 64 times as that of the bulk one.

To study the temperature effect on nanofluidic ionic conductivity, EIS tests were conducted in a constant temperature and humidity box. In contrast with five bulk DESs, the ionic conductivity of GO-DES2, GO-DES4, GO-DES6, GO-DES8, and GO-DES10 membranes are much higher under the whole studied temperatures (Fig. 3d). Take the GO-DES10 system as an example, when the temperature rises from 293 K to 333 K, the ionic conductivity of the GO-confined DES is increased by $715.29 \text{ mS cm}^{-1}$, while the bulk DES is only improved by 7.94 mS cm^{-1} . As shown in Fig. S3, the ionic conductivity of the nanoconfined and bulk DES against different temperatures exhibits a well-fitting Arrhenius-type dependence, from which the activation energy (E_a) can be calculated. All these plots of the ionic conductivity demonstrate the same dependence of temperature for both nanoconfined and bulk DESs. And the influence of temperature becomes more pronounced under nanoconfinement. The conductance of GO-DES2 to GO-DES10 follows an Arrhenius-like behavior, with the E_a of 0.386, 0.325, 0.326, 0.301, and 0.211 eV, respectively (Fig. S4). The corresponding E_a of the bulk DES was 0.150, 0.175, 0.190, 0.185, and 0.172 eV (Fig. S4). Obviously, the E_a of nanoconfined DES is higher than the bulk ones. The increased E_a suggests a relatively strong DES-nanosheets

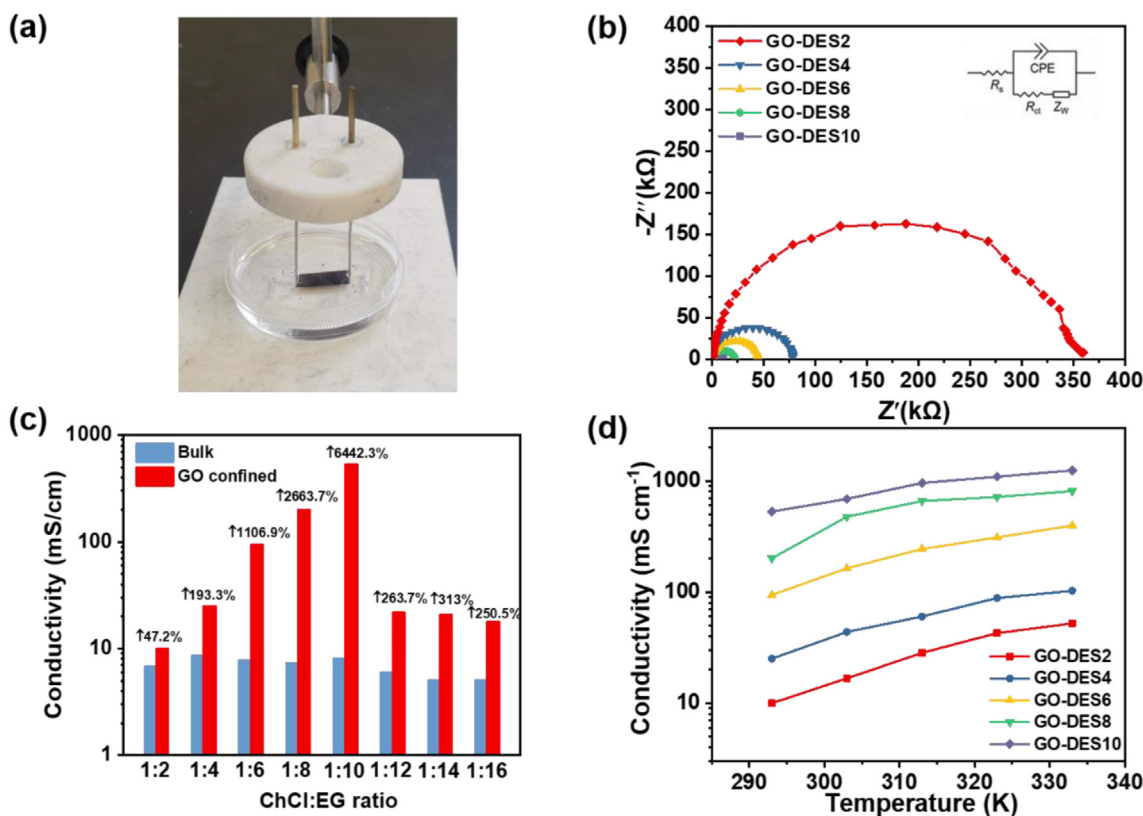


Fig. 3. (a) The nanofluidic device for the electrochemical measurement used in this work. (b) Nyquist plots of nanoconfined DES systems. (c) The increment of the conductivity of the DES due to the nanoconfinement of GO at ambient conditions. (d) The ionic conductivity of nanoconfined systems against temperature. DES, deep eutectic solvents; GO, graphene oxide.

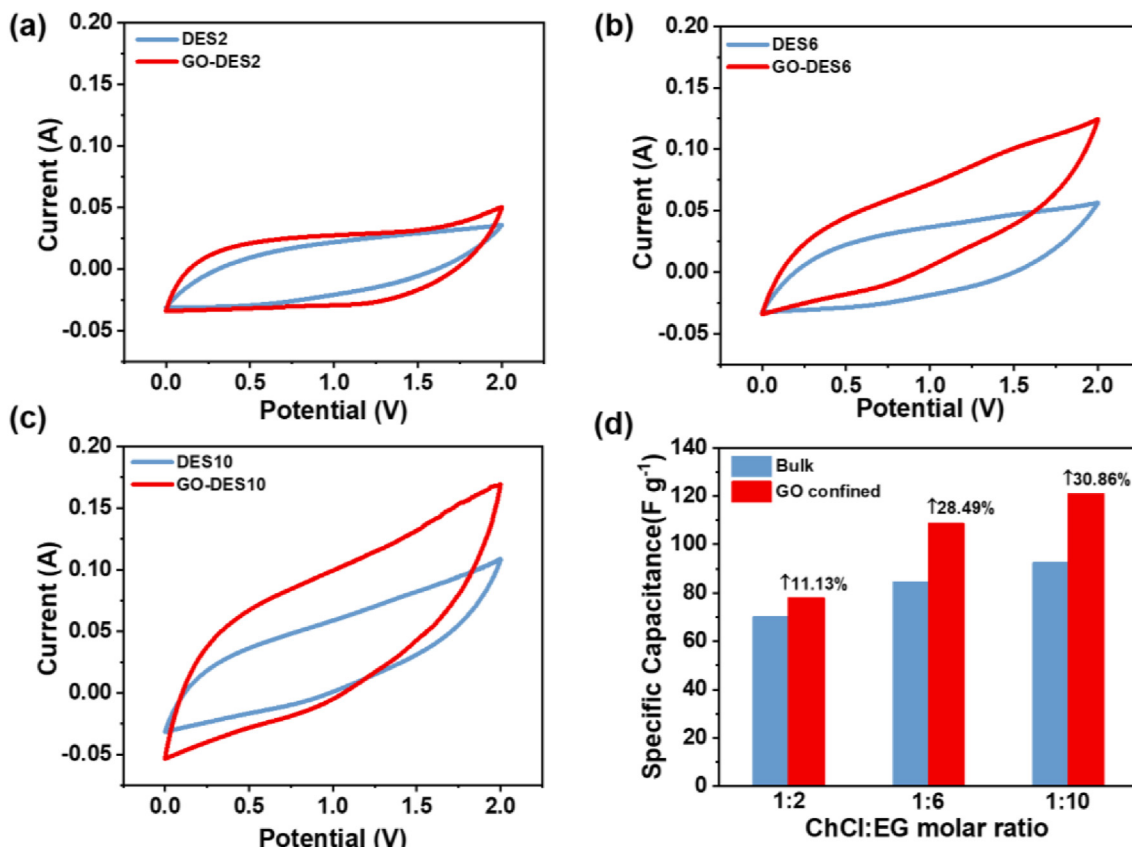


Fig. 4. (a) CV at 50 mV s⁻¹ of the DES2 and GO-DES2 supercapacitor. (b) CV at 50 mV s⁻¹ of the DES 6 and GO-DES6 supercapacitor. (c) CV at 50 mV s⁻¹ of the DES 10 and GO-DES10 supercapacitor. (d) Specific capacitance of different ChCl:EG ratios of the bulk DES supercapacitor and GO-DES supercapacitor at room temperature. DES, deep eutectic solvents; GO, graphene oxide; CV, cyclic voltammetry; EG, ethylene glycol; ChCl, choline chloride.

interaction, which may have benefits on ionic transport of the nanoconfined DES. The low activation energy suggests that protons are transported through GO nanochannels by a Grotthuss mechanism [38]. Among different temperatures, all molar ratios of the nanoconfined DES show higher ionic conductivity than that of the bulk counterparts (Fig. S5), verifying the potential energy storage application. Also, the nanoconfined system maintains relatively high conductivity at high temperatures, which makes it possible that the GO-DES is applied as the electrolyte at high temperatures.

To further understand the nanoconfinement effect of highly increased ionic conductivity, a PFG NMR measurement was carried out to investigate the self-diffusion coefficients of the bulk and nanoconfined DES for a disassociation assessment [39–43]. Fig. S6 shows the ¹H NMR spectrum of ChCl-EG at 298 K, the characteristic resonances of EG ($\delta = 3.05$ ppm for $-\text{CH}_2-$ and $\delta = 4.49$ ppm for $-\text{OH}$) and the resonances of the choline cation. The $-\text{CH}_2-$ resonance at $\delta = 3.05$ ppm of EG overlaps the $-\text{CH}_2-$ resonance of the choline cation. The diffusion coefficients of EG and Ch^+ in the bulk are 5.75×10^{-11} and $3.93 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$ (Table S3). For the GO-DES membrane, the diffusion coefficients of EG and Ch^+ are 1.30×10^{-10} and $1.73 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ (Table S4). The almost twice improved diffusion coefficients confirm the promoting effect of nanoconfinement on ion transport of the nanoconfined DES. The negatively charged GO nanosheet attracts ionized positively charged choline ions. EG molecules and chloride ions are confined in the 2D nanochannel to form more regular and ordered ion transport channels (Fig. 1). Also, the more regular ion transport channel constructed by GO is conducive to ultra-fast ion transport [23,24]. The charge carriers of the supercapacitor are protons, Ch^+ , and negatively charged Cl^- . One part is protons forming hydrogen

bonds inside the DES, another part is unbonded protons. Protons in the hydrogen bond formed between the donor (EG) and acceptor (ChCl) can transfer toward the corresponding electrode surface. Meanwhile, free protons can form hydrogen bonds with the oxygen-containing functional groups ($-\text{COOH}$ or $-\text{OH}$) on the surface of AC, which also accelerates the transport of electrolyte ions and thus achieves higher electrical conductivity. Transport numbers, representing the relative contribution of the specific charged species to the total ionic conductivity [21], are another essential parameter when evaluating the transport behaviors of ions, and they can be calculated from equation (6)

$$t^{+/-} = \frac{D_{\text{cation/anion}}}{D_{\text{cation}} + D_{\text{anion}}} \quad (6)$$

According to the diffusion coefficients obtained from the NMR DOSY measurement, the cationic transport numbers of the bulk and GO-confined are calculated to be 0.49 and 0.46, respectively, while the anionic transport numbers are 0.51 and 0.54, respectively. The gradual leveling of cationic and anionic transport numbers under the nanoconfinement state may be due to negatively charged nanosheets of GO, which adsorb the cation on the channel surface and accelerate the anion motion.

3.3. Electrochemical performance of DES and GO-DES supercapacitor

To expand the application prospect of nanoconfinement in the electrochemical field, DES and GO-DES supercapacitors were conducted. To explore the effect of nanoconfinement on

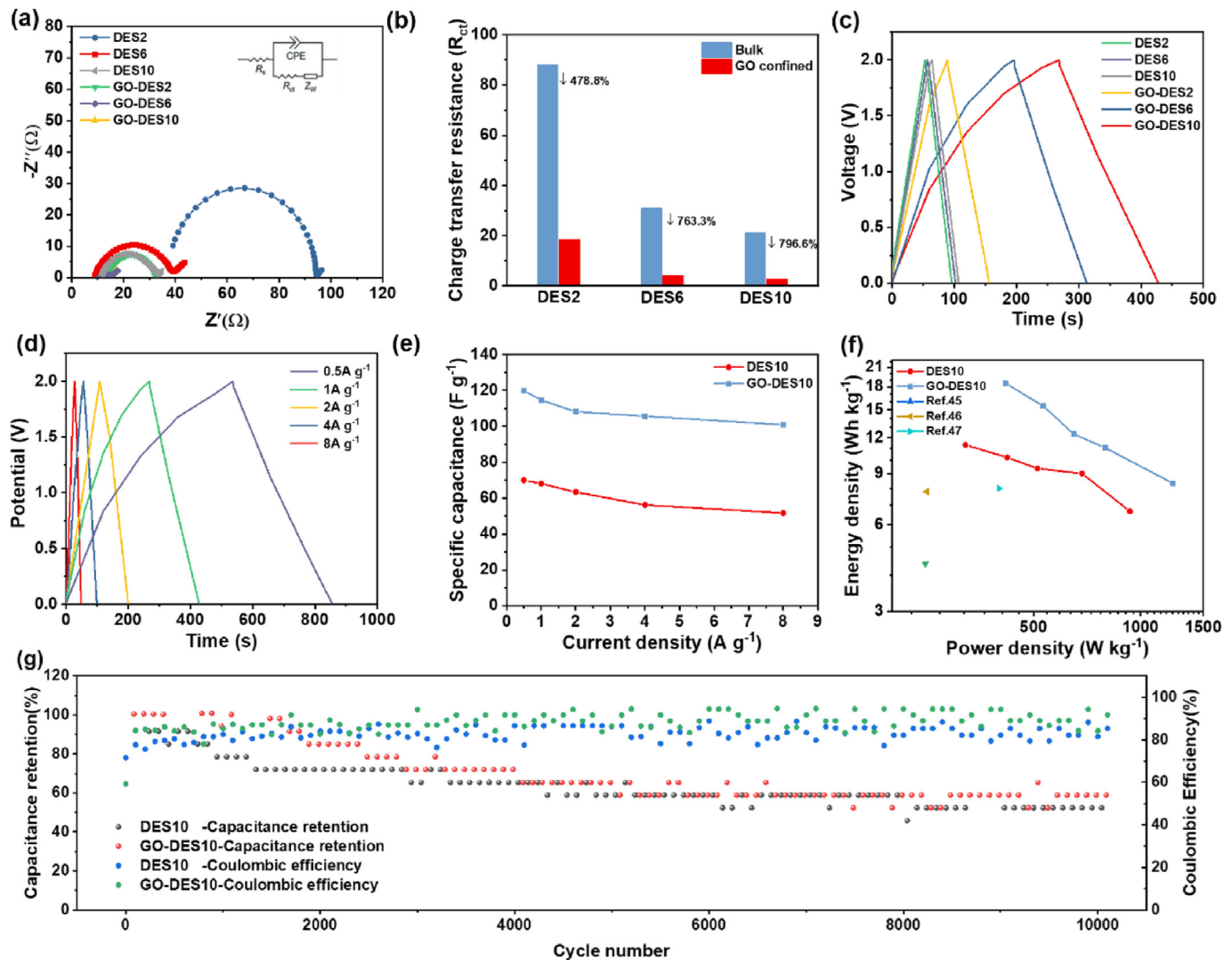


Fig. 5. (a) Nyquist plots of the DES2, DES6, DES10, GO-DES2, GO-DES6, and GO-DES10 supercapacitor at room temperature. (b) Charge transfer resistance (R_{ct}) of the DES2, DES6, and DES10, and their nanoconfined supercapacitor. (c) GCD at 1 A g^{-1} of the DES2, DES6, DES10, GO-DES2, GO-DES6, and GO-DES10 supercapacitor. (d) GCD at $0.5\text{--}8 \text{ A g}^{-1}$ of the GO-DES10 supercapacitor. (e) Specific capacitance vs. current density of the DES10 and GO-DES10 supercapacitor. (f) Ragone diagram. (g) Cyclic charge-discharge tests of the capacitance retention and coulombic efficiency of DES10 and GO-DES10 systems at the current density of 1 A g^{-1} . GCD, galvanostatic charge and discharge; DES, deep eutectic solvents; GO, graphene oxide.

electrochemical properties, CV, GCD, and EIS of DES and GO-DES supercapacitors were evaluated using a two-electrode device (button batteries using AC as the working electrode, Fig. S7) under room temperature. As revealed in Fig. S8, the CV curves of the DES and GO-DES exhibit a nearly quasi-rectangular shape at $50\text{--}250 \text{ mV s}^{-1}$, manifesting that the dominant capacitance is contributed by an electric double layer capacitor. From the CV curves at 50 mV s^{-1} (Fig. 4), the sample of GO-DES2, GO-DES6, and GO-DES10 has larger closed areas than their bulk electrolytes. The specific capacitance of the electric double layer capacitor, calculated from the CV results, are 70.01 (DES2), 77.8 (GO-DES2), 84.39 (DES6), 108.43 (GO-DES6), 92.31 (DES10), and 120.8 F g^{-1} (GO-DES10) at a scanning rate of 50 mV s^{-1} , respectively. Comparing with the bulk counterparts, the promotion rates of GO-DES2, GO-DES6, and GO-DES10 are 11.13% , 28.49% , and 30.86% , respectively. The CV tests of bulk and nanoconfined supercapacitors at different scanning rates were also conducted. The CV curves (Fig. S8) present a quasi-rectangular shape, which demonstrates a good capacitive

behavior of the system. Compared with bulk supercapacitors, the capacity of the supercapacitor in the nanoconfined system is significantly increased, which is consistent with the results obtained from the nanoconfined conductivity test of nanofluids, indicating that a nanoconfined system is good for high performance supercapacitors.

The sample of the nanoconfined DES exhibits better electrochemical performance than the bulk supercapacitor, which matches well with its excellent ionic transport properties. The resistance arising from the electrodes is likely negligible in AC-composed supercapacitors [44]. The main electronic resistance is derived from the interfacial resistance between the inhomogeneity of the electrolyte–electrode interfaces. The supercapacitors with nanoconfinement have smaller charge transfer resistance than the bulk supercapacitors, corresponding to higher ionic conductivity. As is shown in Fig. 5b, compared with the bulk supercapacitors, the charge transfer resistance of nanoconfinement is reduced by 478.8% , 763.3% , and 796.6% , implying that the nanoconfined DES

can effectively improve ionic transport by forming ionic high-speed transport channels to shorten the ionic transport distance. The GCD curves recorded at 1 A g⁻¹ (Fig. 5c) of the bulk and nanoconfined supercapacitors share nearly triangular shapes with slight bending, indicating a very fast electrochemical response during the charge-discharge process. Apparently, nanoconfined supercapacitors, GO-DES2, GO-DES6, and GO-DES10, show larger capacitance of 78 F g⁻¹, 108 F g⁻¹, and 120 F g⁻¹ than DES2, DES6.

DES10 of 70 F g⁻¹, 78 F g⁻¹, and 90 F g⁻¹, respectively. Furthermore, the GCD curves of GO-DES10 recorded at 0.5–8 A g⁻¹ are drawn in Fig. 5d. The unimpressive voltage-drop and symmetric features at high charge-discharge current density demonstrate low equivalent series resistance and high Coulombic efficiency. Consistent with the CV results, GO-DES10 delivers higher capacitance than other samples under coequal current density. Meanwhile, the energy density of GO-DES10 is remarkably better (10.23 Wh kg⁻¹) than that of the DES10 system (6.78 Wh kg⁻¹) at 1 kW kg⁻¹ power density (Fig. 5f). This value surpasses those reported quasi-solid-state electrolytes [45–47]. Most notably, the confined system proposed in this work shows high conductivity, 30.86% specific capacitance improvement, and a 50.88% energy density increment compared with the bulk systems, rendering it one of the promising quasi-solid liquid electrolytes for the future of electrochemical energy storage systems.

In the next stage, the long-term cyclic performance of bulk and nanoconfined supercapacitors was tested to highlight the long-term performance during continuous charge-discharge cycles. Fig. 5g and Fig. S9 demonstrate the variation trend of the capacitance retention vs. cycle number at current density of 1 A g⁻¹. All samples showed a similar trend, including an initial sharp decrease, followed by a stable behavior. Compared with bulk DES supercapacitors, nanoconfined DES showed less capacitance attenuation and relatively better Coulombic efficiency. After the first 2000 cycles, the capacitance retention of GO-DES10 remained 85% higher than DES10 of 73% (Fig. 5g). Moreover, the Coulombic efficiency of GO-DES10 was kept 99.9% higher than DES10 of 90% during the cyclic charging and discharging test. The GO confined system shows higher Coulombic efficiency and capacitance retention than the bulk one, which may be attributed to the higher ionic conductivity and lower transmission resistance providing a more stable ion transport route and less energy loss.

4. Conclusion

In summary, a novel and facile strategy was designed to acquire high ionic transport by the nanoconfinement of DESs into 2D GO nanochannels. The migration of ions in nanoconfined DESs has been proven to be notably accelerated in comparison with bulk DESs. The maximum increment of conductivity by 64 times shows that the nanoconfined system has great significance in the fast ion transportation of DESs. The best performance of the nanoconfined system of ChCl:EG (1:10) demonstrates that an appropriate molar ratio of the hydrogen bond acceptor and donor is important for achieving the highest ionic conductivity. The PFG-NMR proves the promoting effect of nanoconfinement on DESs. At the same time, the assembled supercapacitors show that the nanoconfined system has broad prospects in the electrochemical field. The GO-DES10 exhibits the highest specific capacitance of 120 F g⁻¹ at 1 A g⁻¹ and energy density of 10.23 Wh kg⁻¹ at power density of 1 kW kg⁻¹. The delightful achievement opens a brand-new thought for developing reliable and efficient quasi-solid-state electrolytes. Finding a high conductive and low expenditure nanoconfined DES system could add much value and boost the development of quasi-solid-state electrolytes for supercapacitors and other electronic materials.

Credit author statement

Xiaoyu Wang: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Yuqi Wang:** Conceptualization, Supervision, Writing – review & editing. **Mengyang Dong:** Resources, Writing – review & editing. **Zhou Fang:** Writing – review & editing. **Yue Hu:** Writing – review & editing. **Kainan Xue:** Writing – review & editing. **Zhizhen Ye:** Conceptualization, Funding acquisition, Supervision. **Xinsheng Peng:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtener.2023.101285>.

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